



# Math Talk beyond the Answer Toolkit

## **Pre-K-Grade 12 • Educator and family planning**

Make mathematical reasoning visible through several representations

An original professional toolkit with a five-step routine, rehearsal activities, printable displays, a room-layout planning diagram, observation tools, family pages, and a new-teacher run sheet.

**PRINT • TEACH • REVISIT**



# Choose the version your learners need

## Objective

Make mathematical reasoning visible through several representations. This is an adult planning resource; it is not a state-aligned academic worksheet, an individualized intervention, or a crisis plan.

## Pre-K and Kindergarten

Use one spoken step, a real object or adult demonstration, and a brief supported rehearsal. An adult records observations; children do not need to read the planning forms.

## Grades 1-2

Teach two steps at a time. Pair the printed cue with an example and let learners show or tell what comes next.

## Grades 3-5

Let students explain the sequence, practice roles, and offer one suggestion. Use a short written or drawn reflection.

## Grades 6-8

Offer private feedback and meaningful choices. Ask learners to identify barriers in the routine and help revise one step.

## Grades 9-12

Connect the routine to increasing independence, collaborative projects, and self-advocacy. Invite students to assess whether the system is useful and respectful.



# Teach the routine explicitly

Situation to plan for: Students state numerical answers but cannot connect the operation to the situation.

## 1. Ask what the quantities mean

Describe what students should see or hear. Start with the actual situation: Students state numerical answers but cannot connect the operation to the situation.

## 2. Invite a model or diagram

Model the action instead of only naming it. Keep the cue short: "Invite a model or diagram."

## 3. Connect the model with an equation

Invite a supported rehearsal. Offer a spoken, drawn, or demonstrated response so participation is not limited to writing.

## 4. Compare two methods

Watch one learner or pair try the sequence. If the step is unclear, revise the cue or the environment before repeating it.

## 5. Check the result in context

Ask what helped. Notice a concrete action, then decide whether to keep or adjust the routine.

# Language that teaches the next action

## A usable adult script

What does this number count? Show how your model and equation describe the same situation.

## A response to avoid

Equating speed with mathematical understanding or insisting on one method when several are valid.

## Rehearse with a colleague

Read the situation. Person A uses the script; Person B responds as a learner who needs the first step clarified. Switch roles. Identify the exact words that made the action more understandable.

**Your adapted script, using language natural to your classroom:**

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# Three decision cases

## 1. The first cue does not work

Students state numerical answers but cannot connect the operation to the situation. The adult repeats the same words louder. What information should the adult check, and what could change?

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## 2. The routine works for some learners

Most students use the routine, but one learner needs a different response method. How can the goal stay consistent while access changes?

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## 3. Students propose an improvement

A learner says one part of the routine is confusing and suggests a simpler cue. How should the adult respond?

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# Decision-case response guide

## 1. The first cue does not work

Check whether the learner can access the cue, understands the first action, and has the needed materials. Model “Ask what the quantities mean” and invite a supported rehearsal. Increasing volume alone does not clarify the step.

## 2. The routine works for some learners

Keep the goal “make mathematical reasoning visible through several representations.” Offer demonstration, oral response, reduced copying, or another documented support. Compare the actual learning or participation evidence, not whether every learner used an identical method.

## 3. Students propose an improvement

Ask the learner to demonstrate the confusing point. Try a small change to “Connect the model with an equation,” explain any non-negotiable school requirements, and review whether the revised cue helps participation.

Other responses can be appropriate when they preserve dignity, teach a specific action, follow school procedures and individual plans, and include a way to review the result.



# Math Talk beyond the Answer

- 01** Ask what the quantities mean
- 02** Invite a model or diagram
- 03** Connect the model with an equation
- 04** Compare two methods
- 05** Check the result in context



Pause. Find your next step.

## **NOTICE**

What is happening? What do I need to understand?

## **CHOOSE**

Ask what the quantities mean.

## **TRY**

Connect the model with an equation.

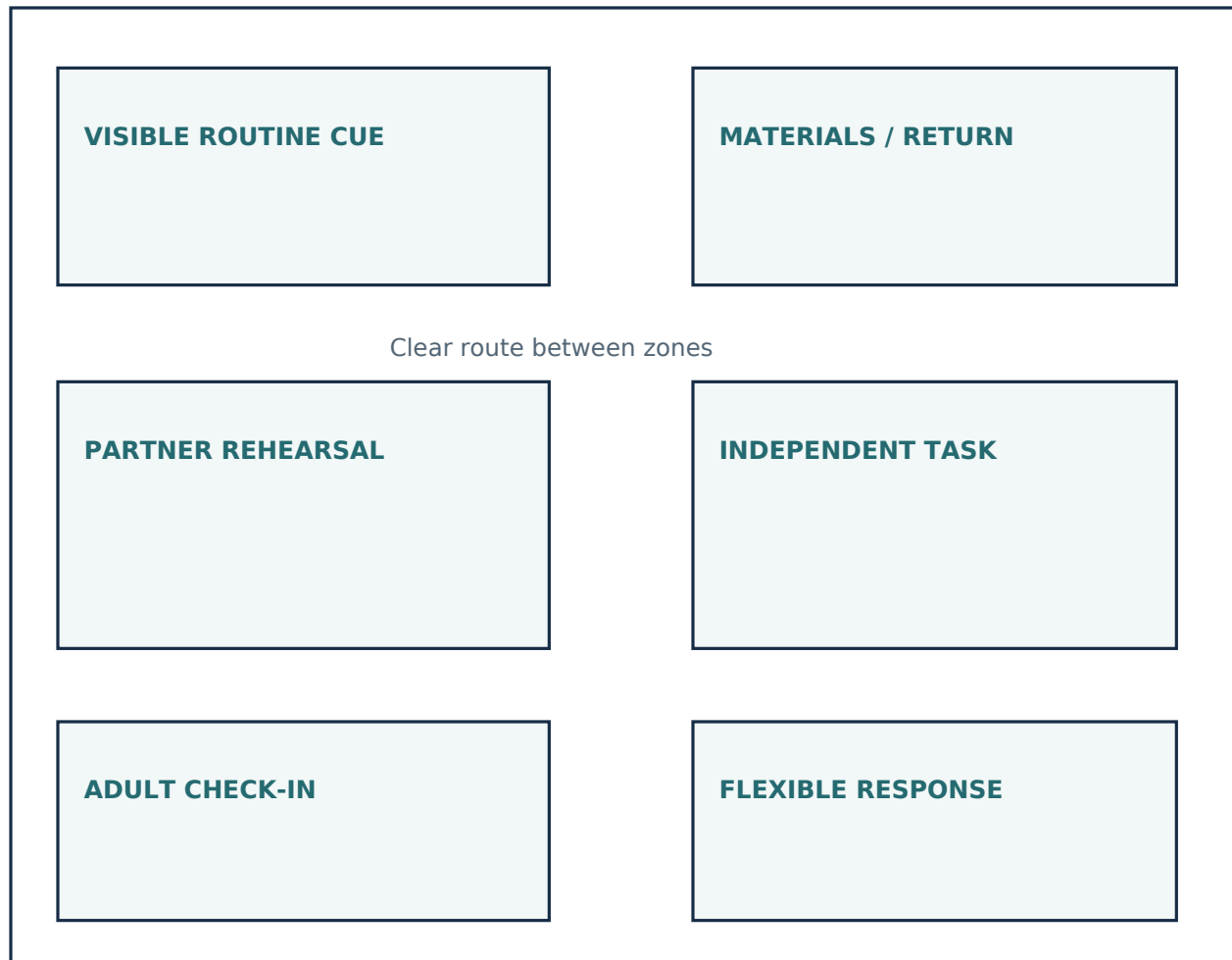
## **CHECK**

Check the result in context.



# A room layout you can adapt

Conceptual layout, not a measured floor plan. Preserve required exits, accessible routes, sight lines, and school rules. The display supports a routine; it does not replace teaching it.



## Display focus: Ask what the quantities mean

Use a routine cue where the action begins. Locate materials so learners can find them. Provide a seated or alternative response route without isolating the learner.

# Make the display earn its space

## Before posting

Print the routine display and the desk cue. Read both from the distance at which students will use them. Enlarge if needed; pair text with real demonstrations for nonreaders.

## Introduce it

Point to the exact step during rehearsal. Ask learners where the cue would be most useful. A poster becomes instructional when students know how to use it.

## Keep it usable

Reduce competing visual clutter near the cue. Keep copies reachable and avoid placing private observation records or student comparisons on display.

## Review after a week

Ask learners to locate the cue and explain its next action. If they cannot use it, revise placement, wording, or instruction instead of adding more posters.

## Visual rights

The typographic posters and functional layout diagram are original Saint Pierre Learning Resources designs. The unchanged KSP logo is the store's supplied brand asset. No stock photography or third-party illustration is included.

# Try the routine in a real activity

## **Activity 1 • Rehearse and revise (15-20 minutes)**

Use a simple equal-sharing situation. Each group represents it with objects, a drawing, and words, then explains how the representations connect.

Prepare blank paper and the printed cue. Model one turn, let partners rehearse, then ask each pair to name one useful step and one improvement. Collect a drawn, oral, or written response.

## **Activity 2 • Teach a new classmate (15 minutes)**

Partners make a short guide for a fictional learner joining the class. Include the purpose “make mathematical reasoning visible through several representations,” the first action, where help is available, and one example of success. Swap guides and test whether the next step is clear.

Success criteria: the guide names an observable action, is understandable to the intended learner, offers a help route, and avoids public judgment of classmates.

# My routine rehearsal

**The routine helps us...**

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**The first action I can show is...**

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**A cue or material that would help me is...**

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**If I am unsure, I can...**

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**One change I would suggest is...**

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# Review what happened

**What action did you observe? Write or draw a concrete example.**

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**Which part was clear, and what evidence supports that view?**

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**What barrier appeared? Consider directions, materials, time, space, or response method.**

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**What one change will you try next, and how will you know whether it helped?**

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# A supportive response plan

## **Describe, then inquire**

Students state numerical answers but cannot connect the operation to the situation. Describe only what you observed. Ask a neutral question about the difficult step before deciding why it happened.

## **Teach the alternative**

What does this number count? Show how your model and equation describe the same situation. Demonstrate the action and provide an opportunity to rehearse with support.

## **Adjust access**

Check the learner's existing plan, task demands, language access, and environment. Choose a support with the learner and relevant staff rather than inventing a diagnosis.

## **Notice progress**

Record a concrete example of participation, such as beginning a step, requesting clarification, or using the cue. Avoid public ranking charts.

## **Review and refer appropriately**

Use the observation form to decide whether the classroom routine needs adjustment. Persistent or safety-related concerns belong with the school's designated support team and procedures.

# Private observation record

**Date / setting / task (avoid unnecessary personal details):**

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**What happened immediately before the observed action?**

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**What did the learner actually say or do?**

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**What cue, model, or access support was offered?**

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**What happened next?**

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**One next instructional adjustment and review date:**

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# Practice neutral observation

**Prompt: Replace “The student did not care” with an observation based on the scenario.**

Possible response: Students state numerical answers but cannot connect the operation to the situation. This names visible actions and the setting without claiming to know the learner’s motives.

**Prompt: What additional information would be useful?**

Ask whether the learner understood the first step, could access the materials or response method, and had the needed support. Note what changes after one specific adjustment.

**Prompt: What would count as improvement?**

A reasonable example is a learner using one step of the routine with a support that works for them, then rejoining the task. Define the observation before collecting it; do not assume fewer visible movements or faster completion alone means better learning.

# A home connection that stays manageable

## Shared purpose

Make mathematical reasoning visible through several representations. The family page supports a short conversation and one manageable routine; families do not need to reproduce the whole classroom system.

## Try this

Ask how the child knows an answer fits a real situation. Accept a drawing or objects before asking for formal notation.

## Conversation opener

“What part feels clear? What part is difficult? Would it help to show me, draw it, or try the first step together?”

## Keep school and home connected

Share a neutral observation and ask which support is already agreed at school. Use the child’s perspective and avoid comparing siblings or classmates.

## If it is not helping

Stop and simplify the routine. Ask the teacher or relevant support professional to help review persistent barriers. This general guide does not replace an individual education, health, or support plan.

# One small step at home

Ask how the child knows an answer fits a real situation. Accept a drawing or objects before asking for formal notation.

**The one step we chose:**

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**What the child said, showed, or asked:**

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**What helped, and what we will adjust:**

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# A five-day family check-in

Use a brief note or drawing. This is not a reward chart or a diagnostic measure. Skip a day if the routine is not relevant.

## Day 1 • What we tried / what helped / next adjustment

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## Day 2 • What we tried / what helped / next adjustment

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## Day 3 • What we tried / what helped / next adjustment

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## Day 4 • What we tried / what helped / next adjustment

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## Day 5 • What we tried / what helped / next adjustment

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# A new teacher's first-week run sheet

## **Before Day 1**

Learn the school's procedures and where individual supports are documented. Choose one routine, not a complete classroom overhaul.

## **Day 1: explain and model**

Use the goal "make mathematical reasoning visible through several representations." Demonstrate "Ask what the quantities mean" and allow a supported rehearsal.

## **Day 2: observe**

Use the private record for one short period. Look for barriers in the task and environment as well as the learner's response.

## **Day 3: ask and adjust**

Invite feedback about "Connect the model with an equation." Change one cue or material, then explain why.

## **Day 4: connect with home**

Use the family page only when relevant. Explain the shared purpose and ask for perspective rather than assigning families a new enforcement role.

## **Day 5: review**

Compare concrete observations. Keep the parts that help, revise a confusing step, and ask a mentor for support when needed.

# Check your plan before using it

**1. Name one observable action the routine teaches.**

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**2. Write the first cue in language your learners can use.**

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**3. Describe an access option that keeps the goal intact.**

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**4. Identify one response to avoid and explain why.**

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**5. Choose one observation and a date to review the routine.**

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# Exit-ticket response guide

1. Ask what the quantities mean. Other concrete actions from the routine are acceptable.
2. What does this number count? Show how your model and equation describe the same situation.
3. Examples include oral response, demonstration, a visible step cue, reduced copying, or a support in the learner's existing plan. Explain how it still serves the goal.
4. Equating speed with mathematical understanding or insisting on one method when several are valid.
5. Choose a specific action in a defined setting, such as use of the first cue during the opening task. Name a realistic review date and the adjustment the observation will inform.

Use this as a planning check. A response is useful when it is specific, respectful, feasible, and connected to evidence. There is no certified score or universal compliance claim.



# Access and responsible use

AIG: Invite learners to evaluate and redesign a routine, compare alternatives, or lead a consent-based demonstration. Do not make advanced learners permanent peer supervisors.

504: Follow the individual plan for pacing, breaks, environment, and response methods. A posted routine does not replace documented accommodations.

EC: Use the IEP and relevant team guidance. Model one step, provide supported practice, and adapt communication and response options.

ESL: Use plain language, demonstrations, home-language discussion, and short sentence frames. Families may respond in their preferred language.

Preparation and printing: Print selected US Letter pages at actual size; enlarge display pages if useful. Forms are printable, not digitally fillable. Keep completed observation and family records private.

Sources and originality: Background reference, accessed September 15, 2026: IRIS Center, Classroom Behavior Management (Part 1), <https://iris.peabody.vanderbilt.edu/module/beh1/>. All scripts, cases, forms, posters, and the layout diagram here are original; no IRIS content or third-party visuals are reproduced.

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